

IT3: Immersive Table Tennis Training Based on 3D Reconstruction of Broadcast Video

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Figure 1: We introduce a virtual reality (VR) table tennis training system named IT3. In Stage 1, we reconstruct the human pose and simulate the ball with AI. Then, we build a system with the following components: (a) Reconstructed Serving Player, (b) Ball Trajectory, (c) Setting of Serve Information, (d) Serve Type Visualization, (e) Movement Visualization of the Player, and (f) Video Player.

Abstract

For professional players, observing their opponents' serves in detail typically involves watching video recordings, usually from broadcast footage. However, broadcast videos often capture players from a side view and lack immersiveness. In this work, we aim to reconstruct 3D table tennis games, incorporating AI-predicted player movements and ball placements, based on any given broadcast table tennis video. Moreover, an editing tool is developed for setting the table position and correcting the wrongly estimated player or ball placement positions. The ball trajectory is then simulated based on the corrected player and ball placement positions. With the proposed simulation system, players can train their reactions and responses to different types of serves in a VR environment from the first-person perspective. They can also observe their opponent's as well as their own movements from any chosen viewing angle, enhancing their training experience.

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1 Introduction

In the rapidly evolving technology field, virtual reality (VR) has proven effective in sports training and education. Table tennis is a fast-paced sport requiring players to react and respond to opponents' movements quickly, and several VR systems have been proposed to enhance table tennis players' motor skills. For example, Oagaz et al. [Oagaz et al. 2021] and Ku et al. [Ku et al. 2022] proposed VR-based table tennis training systems to teach beginners stroke movements. Aiming at teaching beginners how to react to serves with different kinds of spin, SpinPong [Wu et al. 2021] places front-view 2D videos of the opponent in the 3D environment and simulates 3D ball trajectories for different serve types with the visual cues for spin information. However, the ball serving simulation in SpinPong only involves three different trajectories created based on simple rules, lacking diversity. Additionally, SpinPong uses a fixed force to simulate the rebound, failing to mimic complex physical ball movements in real returns.

In this work, targeting coaches, professional players, and enthusiasts, we propose an immersive VR system to enhance practical skills and understanding of tactical and technical elements. We conducted a pilot study involving five professional table tennis players performing various serves recorded with two cameras mounted from the first-person perspective and the side view, respectively. The footage revealed that videos of different views provide various insights into serve skills, and the serve spin type could be identified from the server's detailed swing movements. Therefore, we propose a VR-based table tennis training system that reconstructs the body shapes and movements of the opponent based on a given broadcast table tennis video, allowing users to observe the opponent's movements from different perspectives and practice discriminating the spin type of serves made by a specific opponent. Moreover, our system provides more realistic physical ball simulation, which better models the ball's collisions and spin, thus offering users a training environment that closely resembles real table tennis games.

2 Method

As illustrated in Figure 1, the proposed system is composed of two stages, the first stage aims to reconstruct 3D information of the player and ball from a given broadcast table tennis video, and the second stage visualizes interactive 3D information in VR so that users can observe the player movements and ball trajectories in arbitrary views as well as practice returning the ball in the first-person perspective.

Stage1: 3D Reconstruction And Editing. We use SLAHMR [Ye et al. 2023] to predict players' human poses and reconstruct the human global trajectory in the world coordinate. For 3D ball trajectory, since it is challenging to reconstruct the accurate 3D ball trajectory based on a 2D broadcast video, we instead detect each ball placement and simulate the 3D trajectory of each stroke based on ball physics. The ball placement is obtained by detecting the rebound using a YOLO model and then transforming the 2D rebound position to the 3D world coordinate by applying the homograph transformation matrix, which is estimated according to the four corners of the table. To create a more immersive experience in table tennis simulations, we model the ball's spin effects, including air drag, lift, and collision with spin. Using standard fluid dynamics formulas, we calculate drag and lift coefficients based on a rational function from Schneider et al. [Kemnitz et al. 2013]. For rebound dynamics, we apply Nakashima's [Nakashima et al. 2009] model, incorporating friction and virtual springs to simulate inelastic collisions. To ensure real-time accuracy, we perform calculations in a reference frame where the table and racket are stationary and then project the results back to the original frame. Details of the ball physics are introduced in Section 2 of the Supplementary Materials. In addition, using our developed editor (as shown in Figure 1 of the Supplementary Materials), the 3D positions of the player, the ball placement, and the table can be easily adjusted if the AI-predicted information is inaccurate.

Stage2: User Interaction. Our system provides two modes for user interaction, including a training mode that allows users to practice receiving and returning serves, and an immersive replay mode for watching table tennis matches in a fully immersive environment

(Figure 1). In the training mode, the user selects the type of serve (such as topspin, backspin, sidespin or random) to practice. The opponent's SMPL [Loper et al. 2015] model is estimated based on service clips from broadcast table tennis videos related to the target opponent. The system imparts forward velocity and simulates the spin trajectory, providing a realistic rebound direction upon hitting the table. It also utilizes the Meta Movement SDK [Meta 2022] to track and record users' body poses for review. In immersive replay mode, users can freely explore the match from any angle and use slow motion for detailed analysis, providing better tactical insights.

3 User Study and Future Work

We recruited 17 participants (2 with no table tennis experience, 13 with class learning experience, and 2 with systematic training) to use our system. Each completed a questionnaire on overall ratings and ease of use for the Training and Immersive Replay Modes. See Supplementary Materials for details. In this work, we propose a system to reconstruct a 3D table tennis game from 2D broadcast video based on computer vision technology. We also design a VR user interaction environment for users to observe their opponents' actions in arbitrary views and practice returning the ball. Our experiments revealed that different views are needed to effectively observe the server's habits. However, the AI-based reconstruction of human/ball motions does not always guarantee accuracy, and therefore, it still takes a lot of time to adjust the predicted 3D information. In the future, we plan to collect more data to improve the robustness of 3D estimation AI models and also conduct a more comprehensive user study to explore the effectiveness of using our system.

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